

KAMRUZZAMAN KHAN ALVE

Junior Machine Learning Engineer | Bangla & Multilingual NLP | Transformers

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PROFILE

Recent CSE graduate from RUET (CGPA 3.71/4.00) with two peer-reviewed NLP publications and hands-on experience fine-tuning and evaluating Transformer models for Bangla and multilingual tasks. Built and deployed an XLM-R classifier on Hugging Face, developed a local RAG pipeline, and conducted leakage-safe ML experiments. Strong in Python, PyTorch, Hugging Face Transformers, model benchmarking, reproducible experimentation, and model-failure analysis.

TECHNICAL SKILLS

ML / NLP: Python; PyTorch; Hugging Face Transformers; scikit-learn; XLM-R; BanglaBERT; SahajBERT; mBERT; model fine-tuning; text classification

LLM / Retrieval: LangChain; Ollama; ChromaDB; embeddings; retrieval-augmented generation (RAG); local LLM pipelines

Evaluation / Data: pandas; NumPy; preprocessing; stratified splitting; cross-validation; model benchmarking; error/failure analysis; bootstrap and permutation testing

Engineering: Git; Linux; Jupyter; Gradio; REST APIs; JavaScript/TypeScript; Next.js

RESEARCH & ML EXPERIENCE

Independent NLP Research - Bangla & Multilingual NLP | 2024-Present

- Benchmarked six monolingual and multilingual Transformer encoders on a 25,500-sentence, five-class Bangla dataset; best models reached about 97.6% test accuracy, and the XLM-R model was deployed publicly on Hugging Face.
- Co-developed a unified multilingual vocabulary-difficulty pipeline for the BEA 2026 Shared Task across three L1 groups, emphasizing reproducible model evaluation and cross-language generalization.

Research Member - Young Learners' Research Lab, RUET | 2024-Present

- Conducted AI/NLP research under faculty supervision and participated in collaborative research, seminars, and experiment review across low-resource NLP and biomedical ML.

B.Sc. Thesis - Leakage-Safe EEG Biomarker Discovery for ADHD - RUET CSE | 2025-2026

- Designed subject-level ML evaluation to diagnose data leakage and confounding; showed that random window splitting inflated balanced accuracy by 16.5 points versus honest subject-level validation, then used nested CV and statistical resampling for robust analysis.

Industrial Trainee - Code Studio | 2025

- Worked with requirements analysis, debugging, Git-based collaboration, documentation, and delivery-focused software-development workflows.

SELECTED PROJECTS

Bangla Sentence Classifier | [Hugging Face](#) - Fine-tuned XLM-R with PyTorch/Transformers for five Bangla grammatical sentence types and deployed the model with a Gradio interface.

Web-RAG | [GitHub](#) - Built a local webpage-retrieval and offline LLM question-answering pipeline using Python, LangChain, Ollama, and ChromaDB.

Resume Evaluator | [GitHub](#) - Built an LLM-assisted resume-analysis app with PDF/DOCX parsing, ATS scoring, targeted feedback, in-browser editing, and document export.

PUBLICATIONS

1. **K. K. Alve et al.** "Monolingual vs. Multilingual BERT: Analyzing Performance on Downstream Grammatical Sentence-Type Classification Task." IEEE QPAIN 2026. [DOI](#)
2. **A. A. Hossain and K. K. Alve.** "Failure at BEA 2026 Shared Task 1: One Pipeline, Three L1s: A Unified Language-Agnostic System for Vocabulary Difficulty Prediction." BEA 2026. [ACL Anthology](#)

EDUCATION

B.Sc. in Computer Science and Engineering - Rajshahi University of Engineering & Technology (RUET) | 2022-2026 | CGPA: 3.71/4.00